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Inventor(s)

Uzelac; Nenad et al.

MAGNETIC ACTUATOR

Abstract

A magnetic actuator may include a stator supporting a plurality of permanent magnets and a coil. The magnetic actuator may further include a plunger assembly movable along an axis relative to the stator between a first position and a second position in response to energizing the coil. The plunger assembly may be configured to move at least 1.0 cm along the axis between the first position and the second position in 6 ms or less in response to energizing the coil.

Inventors: Uzelac; Nenad (St. John, IN), Stojkovic; Milos (Nis, RS)

Applicant: G & W Electric Company (Bolingbrook, IL)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/557,060, filed Feb. 23, 2024, the entire content of which is incorporated herein by reference.

FIELD

[0002] The present invention relates to magnetic actuators for electrical switchgear.

BACKGROUND

[0003] Electrical switchgear may include a movable contact that is movable between an open position and a closed position. Electrical switchgear used in power transmission and distribution systems may be energized with relatively high voltage, such as 15.5 kV, 27 kV, 38 kV, 72.5 kV, etc.

SUMMARY

[0004] When the electrical switchgear is energized and the contacts are separated, arcing may occur between the contacts during the initial moments of separation. It is desirable to separate the contacts as quickly as possible to minimize arcing. It is also desirable to separate the contacts as quickly as possible to stop the flow of electrical current through the switchgear in response to a detected fault. By stopping the flow of electrical current quickly in response to a fault, adverse effects of the fault on electrical equipment and property may be minimized.

[0005] Thus, a need exists for an actuator for electrical switchgear that can separate the contacts more quickly than known actuators.

[0006] In some aspects, the techniques described herein relate to a magnetic actuator for a switchgear system, the magnetic actuator including: a stator supporting a plurality of permanent magnets and a coil; and a plunger assembly movable along an axis relative to the stator between a first position and a second position in response to energizing the coil, wherein the plunger assembly is configured to move at least 1.0 cm along the axis between the first position and the second position in 6 ms or less in response to energizing the coil.

[0007] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the stator includes a cylindrical frame.

[0008] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the plunger assembly includes a shaft extending along the axis and a plunger body fixed to the shaft, and wherein the plunger body has a mass less than 1 kg.

[0009] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the plunger body includes a tapered portion.

[0010] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the plunger body includes a first end, a second end opposite the first end, a first portion extending from the first end, and a second portion extending from the second end, wherein the first portion includes the tapered portion, and wherein the second portion is cylindrical.

[0011] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the plunger body includes a third portion between the first portion and the second portion, wherein the third portion is cylindrical, and wherein an outer diameter of the third portion is greater than an outer diameter of the second portion to define a lip.

[0012] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the first end of the plunger body engages the stator when the plunger assembly is in the first position, and wherein the lip engages the stator when the plunger assembly is in the second position.

[0013] In some aspects, the techniques described herein relate to a magnetic actuator, further including a spring biasing the plunger assembly toward the second position.

[0014] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the spring has a spring constant of at least 30 N/mm.

[0015] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the plunger assembly includes a plunger body having a recess, and wherein the spring is received within the recess.

[0016] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the recess is tapered.

[0017] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the stator includes a frame surrounding the coil, and wherein the frame is cylindrical.

[0018] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the coil is a first coil, and wherein the magnetic actuator further includes a second coil supported by the stator.

[0019] In some aspects, the techniques described herein relate to a magnetic actuator, further including a spring biasing the plunger assembly in a first direction toward the second position, and wherein the second coil is configured to be energized to produce a force on the plunger assembly in the first direction.

[0020] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the first coil and the second coil are configured to be energized to move the plunger assembly from the first position toward the second position.

[0021] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the plurality of permanent magnets is configured to retain the plunger assembly in the first position.

[0022] In some aspects, the techniques described herein relate to a magnetic actuator for a switchgear system, the magnetic actuator including: a stator supporting a plurality of permanent magnets and a coil; a plunger assembly movable along an axis relative to the stator between a first position and a second position in response to energizing the coil, wherein the plunger assembly includes a shaft extending along the axis and a plunger body fixed to the shaft, the plunger body including a first end, a second end opposite the first end, a first portion extending from the first end, the first portion having a tapered shape, a second portion extending from the second end, the second portion having a cylindrical shape, and a third portion between the first portion and the second portion, wherein the third portion is cylindrical, and wherein an outer diameter of the third portion is greater than an outer diameter of the second portion to define a lip; a spring biasing the plunger assembly in a first direction toward the second position; and a permanent magnet configured to retain the plunger assembly in the first position, wherein the first end of the plunger body engages the stator when the plunger assembly is in the first position, and wherein the lip engages the stator when the plunger assembly is in the second position.

[0023] In some aspects, the techniques described herein relate to a magnetic actuator for a switchgear system, the magnetic actuator including: a stator supporting a plurality of permanent magnets, a first coil, and a second coil; a plunger assembly movable along an axis relative to the stator from a first position toward a second position in response to energizing the first coil and the second coil; and a spring configured to bias the plunger assembly toward the second position, wherein the plurality of permanent magnets is configured to retain the plunger assembly in the first position via a magnetic attractive force greater than a biasing force of the spring, and wherein the first coil and the second coil, when energized, produce magnetic forces on the plunger assembly, which, together with the biasing force of the spring, overcome the magnetic attractive force to move the plunger assembly to the second position.

[0024] In some aspects, the techniques described herein relate to a magnetic actuator, wherein the stator includes a frame surrounding the first coil, and wherein the frame is cylindrical.

[0025] In some aspects, the techniques described herein relate to a magnetic actuator, further including a first end cap coupled to the frame and a second end cap coupled to the frame opposite the first end cap, wherein the second end cap is positioned between the frame and the second end cap.

[0026] Other features and aspects of the disclosure will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a perspective view of a magnetic actuator for a switchgear system according to an embodiment of the present disclosure.

[0028] FIG. 2 is a cross-sectional perspective view of the magnetic actuator of FIG. 1 taken along line 2-2 in FIG. 1.

[0029] FIG. 3A is a cross-sectional view of the magnetic actuator of FIG. 1 taken along line 3-3 in FIG. 1, showing a first magnetic field generated by an energized first coil.

[0030] FIG. 3B is a cross-sectional view of the magnetic actuator of FIG. 1 taken along line 3-3 in FIG. 1, showing a second magnetic field generated by energizing the first coil.

[0031] FIG. 4 is a cross-sectional view of the magnetic actuator of FIG. 1, taken along line 2-2 in FIG. 1, showing airflow pathways.

[0032] FIG. 5 is a cross-sectional view of the magnetic actuator of FIG. 1, taken along line 5-5 in FIG. 1, showing airflow pathways.

[0033] FIG. 6 is a perspective cross-sectional view of a coil of the magnetic actuator of FIG. 1.

[0034] FIG. 7 is a perspective cross-sectional view of a plunger body of the magnetic actuator of FIG. 1.

[0035] FIG. 8 is a cross-sectional view of the plunger body of the magnetic actuator of FIG. 1.

[0036] FIG. 9 is an upper perspective view of an outer portion of a first end cap of a stator of the magnetic actuator of FIG. 1.

[0037] FIG. 10 is a lower perspective view of the outer portion of FIG. 9.

[0038] FIG. 11 is a perspective view of an inner portion of the first end cap of the stator of the magnetic actuator of FIG. 1.

[0039] FIG. 12 is a perspective view of a second end cap of the stator of the magnetic actuator of FIG. 1.

[0040] FIG. 13 is a perspective view of a bearing support of the magnetic actuator of FIG. 1.

[0041] FIG. 14 is a perspective view of an end plate of the magnetic actuator of FIG. 1.

[0042] FIG. 15 is a partial exploded view of the magnetic actuator of FIG. 1.

[0043] Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.

[0044] Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. For example, the use of directional terms such as upper, lower, top, bottom, left, and right are used for descriptive purposes only with reference to the particular orientations illustrated in the figures.

DETAILED DESCRIPTION

[0045] FIG. 1 illustrates a magnetic actuator 10 for a switchgear system according to an embodiment of the present disclosure. The magnetic actuator 10 may be coupled, for example, to a movable contact of a vacuum interrupter or other loadbreak device of the switchgear system, and is operable to move the movable contact between an open position and a closed position.

[0046] With continued reference to FIG. 1, the magnetic actuator 10 includes a stator 14 having a cylindrical frame 18, a first end cap 22 positioned on an upper (first) side of the frame 18, and a second end cap 26 positioned on a lower (second) side of the frame 18. The frame 18 defines a first longitudinal axis A1 extending centrally through the frame 18 and the end caps 22, 26.

[0047] Referring to FIG. 2, the first end cap 22 includes an outer portion 38 and an inner portion 42. The outer portion 38 and the inner portion 42 of the first end cap 22 may be made of a

ferromagnetic material, such as steel. A bearing support **30** is received by the inner portion **42**. The bearing support **30** may be made of non-magnetic material, such as aluminum. The illustrated bearing support **30** includes a tapered wall **212** that is tapered inwardly in a direction moving from an upper (i.e., first) side **214** of the bearing support **30** toward a lower (i.e., second) side **218** of the bearing support **30** (FIG. 13).

[0048] A first linear bearing **150** is positioned within the bearing support **30** to support a shaft **34**, such that the shaft **34** is slidable along the axis **A1** (FIG. 2). In the illustrated embodiment, the shaft **34** is further supported by a second linear bearing **151**, which in turn is supported by the second end cap **26**. The linear bearing **150** may be a roller bearing, a bushing, or the like. The shaft **34** is configured to be coupled to the movable contact (e.g., through an insulating operating rod), such that longitudinal motion of the shaft **34** may be transferred to the movable contact to move the contact between the open and closed positions.

[0049] With continued reference to FIG. 2, a plunger body **46** is fixed (e.g., by press-fitting, welding, etc.) to the shaft **34** for movement therewith along the longitudinal axis **A1**. The plunger body **46** includes a first end **46a** closest to and facing the first end cap **22** and a second end **46b** opposite the first end **46a**. Best illustrated in FIG. 7, a first portion **50** of the plunger body **46** extends from the first end **46a** toward the second end **46b**, and a second portion **54** extends from the second end **46b** toward the first end **46a**. In the illustrated embodiment, the first portion **50** is at least partially tapered or frusto-conical, such that at least part of the first portion **50** has an outer surface **50a** with an outer diameter that increases in a direction along the longitudinal axis **A1** from the second end **46b** toward the first end **46a**. An inner surface **50b** of the first portion **50** is also at least partially tapered or frusto-conical, such that a frusto-conical recess **51** is defined within the first portion **50** of the plunger body **46**.

[0050] With continued reference to FIG. 7, the second portion **54** of the plunger body **46** is cylindrical in the illustrated embodiment, with an outer diameter that is constant along the longitudinal axis **A1**. In the illustrated embodiment, the plunger body **46** further includes a third portion **58** between the first portion **50** and the second portion **54**. The third portion **58** is cylindrical in the illustrated embodiment, with an outer diameter that is constant along the longitudinal axis **A1**, but that is larger than the outer diameter of the second portion **54**, such that a lip or shoulder **59** is located at an interface between the second portion **54** and the third portion **58**. In the illustrated embodiment, the recess **51** also includes a cylindrical portion that extends centrally along the longitudinal axis **A1** into the third portion **58** and the second portion **54** of the plunger body **46**.

[0051] With reference to FIG. 7, the plunger body **46** includes a two-dimensional cross-sectional projection having a geometry that changes as the cross-sectional projection moves along the axis **A1**. In the illustrated embodiment, the inner surface **50b** extends outwardly from the longitudinal axis **A1** at a conical angle α of approximately 15 degrees (FIG. 8). In some embodiments, the conical angle α may be between 1 degree and 89 degrees, between 5 degrees and 30 degrees, or the like. The outer surface **50a** extends outwardly from the longitudinal axis **A1** at the conical angle α . The inner surface **50b** is defined by a variable radius **R1** measured from the longitudinal axis **A1**. The outer surface **50a** is defined by a variable radius **R2** measured from the longitudinal axis **A1**. A difference **X** between the variable radius **R2** and the variable radius **R1** at any particular point along the longitudinal axis **A1** is constant over at least a part of the first portion **50** of the plunger body **46**. Due to this geometry, the area of the cross-sectional projection is sufficiently large at all points along the axis **A1** to avoid magnetic saturation of the plunger body **46**.

[0052] The plunger body **46** and the shaft **34** together form a plunger assembly **60** that is movable relative to the stator **14** between a first position (shown in FIG. 3A) corresponding to the closed position of the moveable contact, and a second position (shown in FIG. 3B) corresponding to the open position of the movable contact. As described in greater detail below, the contoured shape of the plunger body **46** reduces the mass of the plunger body **46** compared to, for example, a

completely cylindrical plunger body. In the illustrated embodiment, the plunger body **46** advantageously has a mass that is less than 1 kg, such that the plunger body **46** has a relatively low inertia and can accelerate quickly, thereby resulting in faster movement of the movable contact and reduced arcing between the movable contact and the fixed contact. The contoured shape of the plunger body **46** may also advantageously minimize magnetic field energy losses and avoid saturation.

[0053] In the first (contacts closed) position of the plunger assembly **60** (FIG. 3A), the first end **46a** of the plunger body **46** abuts the first end cap **22** and, more specifically, the inner portion **42** of the first end cap **22**. In the second (contacts open) position of the plunger assembly **60** (FIG. 3B), the first end **46a** of the plunger body **46** is spaced from the first end cap **22**, and the lip **59** of the plunger body **46** abuts a washer **61** supported by the second end cap **26**. Referring again to FIG. 2, a spring **62** is positioned within the recess **51** of the plunger body **46** to bias the plunger body **46** away from the first end cap **22** and toward the second position (i.e., the spring **62** exerts a downward spring force **71a** against the plunger body **46**. The spring **62** includes a first end **66** seated against the bearing support **30** and a second end **70** seated against a bottom end of the recess **51**. The spring **62** may have a relatively high spring constant of (e.g., between 30 and 40 N/mm in some embodiments). In some embodiments, the spring **62** may be formed of a wire with a square or rectangular cross-section, which may contribute to the high spring constant

[0054] With continued reference to FIG. 2, a plurality of permanent magnets **74** is supported by the first end cap **22**, between the outer portion **38** and the inner portion **42** of the first end cap **22**. The magnets **74** may include, for example, neodymium or other rare earth materials. The magnets **74** are circumferentially positioned about the longitudinal axis **A1** in a generally octagonal shape. The magnets **74** generate a magnetic field extending through the inner portion **42** of the first end cap **22** and into the plunger body **46** when the plunger assembly **60** is in its first position, developing a magnetic attractive force **86** that retains the plunger assembly **60** in the first position against the spring force **71a**. That is, the magnetic attractive force **86** is greater in magnitude than the spring force **71a** in some embodiments, or at least greater than a sum of the spring force **71a** and any gravitational force acting in the direction of the spring force **71a** (e.g., the weight of the plunger assembly **60**).

[0055] With reference to FIGS. 2 and 6, a main coil **90** is positioned around the plunger body **46**. The main coil **90** is partially surrounded by a coil frame **142**. In the illustrated embodiment, the main coil **90** is surrounded on three sides by the coil frame **142**, such that the coil frame **142** has a U-shaped cross-section. The coil frame **142** is positioned between the main coil **90** and the plunger body **46**, between the main coil **90** and the first end cap **22**, and between the main coil **90** and the second end cap **26**. The main coil **90** includes a wire **91** wrapped continuously around the coil frame **142** to a thickness, for example, of five layers of wire **91** in the illustrated embodiment. In other embodiments, different numbers of layers of wire **91** may be used. The illustrated wire **91** is made of a conductive material, such as copper. Free ends **92a**, **92b** (FIG. 6) of the wire **91** are electrically connected to a current source. The configuration of the main coil **90** may provide a relatively low inductance, permitting a rapid increase in current driven through the coil **90**. This may result in the magnetic actuator **10** having a faster operating speed than typical magnetic actuators.

[0056] With reference to FIG. 3B, the main coil **90** may be selectively energized (i.e., by directing current through the wire **91** in a first direction) to generate a first magnetic field **96** that forms a loop around the main coil **90**, represented by the dashed arrows surrounding the main coil **90** in FIG. 3B. The first magnetic field **96** generates a downwardly directed magnetic force **97** on the plunger body **46**. The magnetic force **97** acts in an opposite direction to the magnetic attractive force **86** produced by the permanent magnets **74**. The magnetic force **97** and the spring force **71a** operate together to counteract and overcome the magnetic attractive force **86**, thereby moving the plunger body **46** downwardly from the first (contacts closed) position (FIG. 3A) to the second

(contacts open) position (FIG. 3B).

[0057] Referring to FIG. 3A, the main coil **90** may also be selectively energized (i.e., by directing current through the wire **91** in a second direction opposite the first direction) to generate a second magnetic field **94** that forms a loop around the main coil **90**, represented by the dashed arrows surrounding the main coil **90** in FIG. 3A. The second magnetic field **94** generates an upwardly directed magnetic force **95** on the plunger body **46**. The magnetic force **95** acts in a direction opposite the spring force **71a** and is sufficient to overcome the spring force **71a** (when the spring **62** is in its expanded state corresponding with the second position of the plunger assembly **60**) to move the plunger assembly **60** from the second (contacts open) position (FIG. 3B) toward the first (contacts closed) position (FIG. 3A). As the plunger assembly **60** approaches the first position, the magnetic attractive force **86** grows stronger. Once the plunger assembly **60** reaches the first position (FIG. 3A), the magnetic attractive force **86** is sufficient to maintain the plunger assembly **60** in the first position, and the main coil **90** may be de-energized.

[0058] With reference to FIG. 4, the stator **14** includes a plurality of bores to facilitate assembly of the stator **14**. In particular, the stator **14** includes a first plurality of bores **98** extending between the frame **18** and the outer portion **38** of the first end cap **22**. The first plurality of bores **98** is configured to receive a first plurality of fasteners (not shown) to attach the outer portion **38** of the first end cap **22** to the frame **18**. A second plurality of bores **102** extends between the frame **18** and the second end cap **26**. The second plurality of bores **102** is configured to receive a second plurality of fasteners (not shown) to attach the second end cap **26** to the frame **18**. With returning reference to FIG. 3A, a third plurality of bores **106** extends between an end plate **146** and the outer portion **38** of the first end cap **22**. The third plurality of bores **106** is configured to receive a third plurality of fasteners (not shown) to attach the end plate **146** to the outer portion **38** of the first end cap **22**. With returning reference to FIG. 4, a fourth plurality of bores **110** extends between the end plate **146** and the inner portion **42** of the first end cap **22**. The fourth plurality of bores **110** is configured to receive a fourth plurality of fasteners (not shown) to attach the end plate **146** to the inner portion **42** of the first end cap **22**.

[0059] With reference to FIG. 14, the end plate **146** includes a plurality of mounting bores **222** positioned around the end plate **146** for mounting the end plate **146**, and accordingly the magnetic actuator **10**, to an enclosure of a switchgear system.

[0060] With reference to FIG. 15, a plurality of fasteners **154** fasten the outer portion **38** of the first end cap **22** to the frame **18**. Another plurality of fasteners **156** extend between the end plate **146** and the outer portion **38** of the first end cap **22**. Yet another plurality of fasteners **157** extend between the end plate **146** and the inner portion **42** of the first end cap **22**. The fasteners **154**, **156**, **157** function together to hold the first end cap **22** to the end plate **146**, to support the bearing support **30**, and to support the outer portion **38** and the inner portion **42** of the first end cap such that the magnets **74** are supported between the outer portion **38** and the inner portion **42**.

[0061] The illustrated stator **14** may also include a plurality of bores that define air vents to control the movement speed of the plunger assembly **60**. For example, with reference to FIGS. 4 and 5, a first plurality of air vents **114** is provided in the end plate **146**. The first plurality of air vents **114** may be oriented axially (i.e., parallel to the longitudinal axis **A1**). A second plurality of air vents **118** is provided in the bearing support **30**. The second plurality of air vents **118** may also be oriented axially. A third plurality of air vents **122** is provided in the second end cap **26**. The third plurality of air vents **122** may also be oriented axially.

[0062] A first plurality of lateral air vents **126** is provided perpendicular the longitudinal axis **A1** to vent a first internal cavity **128** between the plunger body **46** and the coil frame **142** to an atmosphere **129**. Each of the first plurality of lateral air vents **126** includes a first portion **126a** between the plunger body **46** and the coil frame **142**, a second portion **126b** between the inner portion **42** and the coil frame **142**, a third portion **126c** between the coil frame **142** and the outer portion **38**, and a fourth portion **126d** between the frame **18** and the outer portion **38**. A second

plurality of lateral air vents **130** is provided perpendicular to the longitudinal axis **A1** to vent the first internal cavity **128** to the atmosphere **129**. Each of the second plurality of lateral air vents **130** includes a first portion **130a** between the plunger body **46** and the coil frame **142**, a second portion **130b** between the coil frame **142** and the second end cap **26**, and a third portion **130c** between the frame **18** and the second end cap **26**. The third plurality of air vents **122** may also vent the first internal cavity **128** to the atmosphere **129**. The first and second pluralities of air vents **114**, **118** may vent a second internal cavity **131**, provided within the conical first portion **50** of the plunger body **46**, to the atmosphere **129**.

[0063] With reference to FIGS. **9** and **10**, the outer portion **38** of the first end cap **22** includes an inside surface **182** having a plurality of flat surfaces **186** that sandwich the magnets **74** between the outer portion **38** and the inner portion **42**. The outer portion **38** includes a lower surface **190** including a plurality of radial channels **194** spaced about the outer portion **38**. Each of the channels **194** forms and/or partially defines part of one of the first plurality of lateral air vents **126**. In the illustrated embodiment, each of the channels **194** forms part of the fourth portion **126d** of one of the first plurality of lateral air vents **126**. The channels **194** are cut into the lower surface **190** and equidistantly spaced about the outer portion **38** of the first end cap **22**.

[0064] With reference to FIG. **11**, the inner portion **42** of the first end cap **22** includes an outer surface **198** having a plurality of flat surfaces **202** that sandwich the magnets **74** between the outer portion **38** and the inner portion **42**. With reference to FIG. **12**, the second end cap **26** includes an upper surface **206** including a plurality of radial channels **210** spaced about the second end cap **26**. Each of the channels **210** forms part of one of the second plurality of lateral air vents **130**. In the illustrated embodiment, each of the channels **210** forms part of the third portion **130c** of one of the second plurality of lateral air vents **130**.

[0065] As shown in FIG. **4**, the air vents allow air displaced by movement of the plunger assembly **60** to escape from between the plunger assembly **60** and the stator **14**. The air vents may also act as restrictive orifices to dampen movement of the plunger assembly **60** and reduce the force of the impact between the plunger assembly **60** and the stator **14** when the plunger assembly **60** reaches the first position and the second position.

[0066] With reference to FIG. **5**, the magnetic actuator **10** may further include a secondary coil **134** is affixed to the stator **14** and more specifically to the second end cap **26**. The secondary coil **134** is a Thomson coil **134** in the illustrated embodiment, and is electrically connected to a current source. The current source may be the same current source or a different current source than the current source electrically connected to the main coil **90**. However, the current directed through the Thomson coil **134** may be greater in magnitude than the current directed through the main coil **90**. For example, in some embodiments, the main coil **90** may be configured to receive 30-100 Amps of current during operation, and the Thomson coil **134** may be configured to receive more than 1,000 Amps of current during operation. A disk **138**, (i.e., a Thomson disk), is affixed (e.g., by press-fitting, welding, etc.) to the plunger body **46** and/or to the shaft **34** for longitudinal motion therewith. As such, the disk **138** may be part of the plunger assembly **60**.

[0067] Energizing the secondary coil **134** produces a magnetic force **158** that acts to separate the disk **138** from the secondary coil **134**. The magnetic force **158** cooperates with the spring force **71a** and the magnetic force **97** to overcome the magnetic attractive force **86** and to move the plunger assembly **60** from the first position (FIGS. **5** and **3A**) to the second position (FIG. **3B**), resulting in faster acceleration of the plunger assembly **60**. A method of operating the magnetic actuator **10** may include energizing the main coil **90** and the secondary coil **134** simultaneously. In other embodiments, the secondary coil **134** may be energized slightly before the main coil **90** is energized, or vice-versa.

[0068] Thus, the present disclosure provides, among other things, a magnetic actuator that may be able to provide a faster actuating speed than known magnetic actuators. For example, in some embodiments, the magnetic actuator **10** provides at least 1.0 cm of travel along the axis **A1** in 6

milliseconds or less. In some embodiments, the magnetic actuator **10** may provide at least 1.0 cm of travel along the axis A1 in 5 milliseconds or less. In some embodiments, the magnetic actuator **10** may provide at least 1.0 cm of travel along the axis A1 in 4 milliseconds or less. These fast actuating speeds, achieved by the constructions of the magnetic actuator **10** described and illustrated herein, may reduce arcing that occurs when the contacts are separated.

[0069] Although the disclosure has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the disclosure as described.

[0070] Various features of the disclosure are set forth in the following claims.

Claims

1. A magnetic actuator for a switchgear system, the magnetic actuator comprising: a stator supporting a plurality of permanent magnets and a coil; and a plunger assembly movable along an axis relative to the stator between a first position and a second position in response to energizing the coil, wherein the plunger assembly is configured to move at least 1.0 cm along the axis between the first position and the second position in 6 ms or less in response to energizing the coil.
2. The magnetic actuator of claim 1, wherein the stator includes a cylindrical frame.
3. The magnetic actuator of claim 1, wherein the plunger assembly includes a shaft extending along the axis and a plunger body fixed to the shaft, and wherein the plunger body has a mass less than 1 kg.
4. The magnetic actuator of claim 3, wherein the plunger body includes a tapered portion.
5. The magnetic actuator of claim 4, wherein the plunger body includes a first end, a second end opposite the first end, a first portion extending from the first end, and a second portion extending from the second end, wherein the first portion includes the tapered portion, and wherein the second portion is cylindrical.
6. The magnetic actuator of claim 5, wherein the plunger body includes a third portion between the first portion and the second portion, wherein the third portion is cylindrical, and wherein an outer diameter of the third portion is greater than an outer diameter of the second portion to define a lip.
7. The magnetic actuator of claim 6, wherein the first end of the plunger body engages the stator when the plunger assembly is in the first position, and wherein the lip engages the stator when the plunger assembly is in the second position.
8. The magnetic actuator of claim 1, further comprising a spring biasing the plunger assembly toward the second position.
9. The magnetic actuator of claim 8, wherein the spring has a spring constant of at least 30 N/mm.
10. The magnetic actuator of claim 8, wherein the plunger assembly includes a plunger body having a recess, and wherein the spring is received within the recess.
11. The magnetic actuator of claim 10, wherein the recess is tapered.
12. The magnetic actuator of claim 1, wherein the stator includes a frame surrounding the coil, and wherein the frame is cylindrical.
13. The magnetic actuator of claim 1, wherein the coil is a first coil, and wherein the magnetic actuator further comprises a second coil supported by the stator.
14. The magnetic actuator of claim 13, further comprising a spring biasing the plunger assembly in a first direction toward the second position, and wherein the second coil is configured to be energized to produce a force on the plunger assembly in the first direction.
15. The magnetic actuator of claim 14, wherein the first coil and the second coil are configured to be energized to move the plunger assembly from the first position toward the second position.
16. The magnetic actuator of claim 15, wherein the plurality of permanent magnets is configured to retain the plunger assembly in the first position.
17. A magnetic actuator for a switchgear system, the magnetic actuator comprising: a stator

supporting a plurality of permanent magnets and a coil; a plunger assembly movable along an axis relative to the stator between a first position and a second position in response to energizing the coil, wherein the plunger assembly includes a shaft extending along the axis and a plunger body fixed to the shaft, the plunger body including a first end, a second end opposite the first end, a first portion extending from the first end, the first portion having a tapered shape, a second portion extending from the second end, the second portion having a cylindrical shape, and a third portion between the first portion and the second portion, wherein the third portion is cylindrical, and wherein an outer diameter of the third portion is greater than an outer diameter of the second portion to define a lip; a spring biasing the plunger assembly in a first direction toward the second position; and a permanent magnet configured to retain the plunger assembly in the first position, wherein the first end of the plunger body engages the stator when the plunger assembly is in the first position, and wherein the lip engages the stator when the plunger assembly is in the second position.

18. A magnetic actuator for a switchgear system, the magnetic actuator comprising: a stator supporting a plurality of permanent magnets, a first coil, and a second coil; a plunger assembly movable along an axis relative to the stator from a first position toward a second position in response to energizing the first coil and the second coil; and a spring configured to bias the plunger assembly toward the second position, wherein the plurality of permanent magnets is configured to retain the plunger assembly in the first position via a magnetic attractive force greater than a biasing force of the spring, and wherein the first coil and the second coil, when energized, produce magnetic forces on the plunger assembly, which, together with the biasing force of the spring, overcome the magnetic attractive force to move the plunger assembly to the second position.

19. The magnetic actuator of claim 18, wherein the stator includes a frame surrounding the first coil, and wherein the frame is cylindrical.

20. The magnetic actuator of claim 19, further comprising a first end cap coupled to the frame and a second end cap coupled to the frame opposite the first end cap, wherein the second end cap is positioned between the frame and the second end cap.
